In [47]:
```
# Example Dataset one
import pandas as pd

# Read the dataset
data_orignal = pd.read_csv("Dataset/Airbnb_dataset/Airbnb_Open_Data.csv")

# Display first five rows
#display(data_orignal.head())
df1 = data_orignal.copy()
```

```
C:\Users\himats\AppData\Local\Temp\ipykernel_27336\1181998762.py:5: DtypeWarning:
Columns (25: license) have mixed types. Specify dtype option on import or set low
_memory=False.
  data_orignal = pd.read_csv("Dataset/Airbnb_dataset/Airbnb_Open_Data.csv")
```

In [ ]:
```
df1 = data_orignal.copy()
# Display records
display(df1.head())
display(df1.head(10))
display(df1.tail())
display(df1.sample(5))
```

In [ ]:
```
# Dataset dimensions
print(df1.shape)
print(len(df1))

# Column names
print(df1.columns.tolist())

# Data types and information
print(df1.dtypes)
df1.info()

# Statistical summary
display(df1.describe())
display(df1.describe(include="all").T)
```

Watch youtube video and details : https://www.youtube.com/watch?v=I7DZP4rVQOU&list=PLTsu3dft3CWhOUPyXdLw8DGy_1l2oK1y

Step 1: Deleting Redundant Columns

Some datasets contain columns that are unnecessary for a particular analysis.

Removing these columns makes the dataset easier to understand and process.

Pandas provides `drop()` to remove columns.

We should only remove columns that are not needed for our analysis.

In [ ]:
```
```

There are two ways

(1) Just drop columns using names (2) Filter out columns

In [48]:
```
Columns_to_keep =  [ 'NAME', 'host id', 'host_identity_verified', 'host name',
       'neighbourhood group', 'neighbourhood', 'lat', 'long', 'country',
       'country code', 'instant_bookable', 'cancellation_policy', 'room type',
       'Construction year', 'price', 'service fee', 'minimum nights',
       'number of reviews', 'last review']
columns_to_drop = ['reviews per month',
       'review rate number', 'calculated host listings count',
       'availability 365', 'house_rules', 'license', 'id']

print(len(columns_to_drop))
print(len(Columns_to_keep))
```

```
7
19
```

In [49]:
```
df1.columns
```

Out[49]:
```
Index(['id', 'NAME', 'host id', 'host_identity_verified', 'host name',
       'neighbourhood group', 'neighbourhood', 'lat', 'long', 'country',
       'country code', 'instant_bookable', 'cancellation_policy', 'room type',
       'Construction year', 'price', 'service fee', 'minimum nights',
       'number of reviews', 'last review', 'reviews per month',
       'review rate number', 'calculated host listings count',
       'availability 365', 'house_rules', 'license'],
      dtype='str')
```

In [51]:
```
# (1) Filtering data

df1 = df1[Columns_to_keep]
```

In [52]:
```
df1.shape
```

Out[52]:
```
(102599, 19)
```

In [53]:
```
df2 = data_orignal.copy()
df2.shape
```

Out[53]:
```
(102599, 26)
```

In [60]:
```
# Second method using drop columns
#df2.drop(columns = ["license"])
df2.shape
```

Out[60]:
```
(102599, 25)
```

In [67]:
```
#df3 = data_orignal.copy()
#df3 = df3.drop(columns =columns_to_drop, inplace = True)
```

Task 2 Rename the columns

In [69]:
```
df3 = data_orignal.copy()
df3.columns
```

Out[69]:
```
Index(['id', 'NAME', 'host id', 'host_identity_verified', 'host name',
       'neighbourhood group', 'neighbourhood', 'lat', 'long', 'country',
       'country code', 'instant_bookable', 'cancellation_policy', 'room type',
       'Construction year', 'price', 'service fee', 'minimum nights',
       'number of reviews', 'last review', 'reviews per month',
       'review rate number', 'calculated host listings count',
       'availability 365', 'house_rules', 'license'],
      dtype='str')
```

In [72]:
```
df3.rename(columns ={"NAME": "Name"}, inplace=True)
df3.columns
```

Out[72]:
```
Index(['id', 'Name', 'host id', 'host_identity_verified', 'host name',
       'neighbourhood group', 'neighbourhood', 'lat', 'long', 'country',
       'country code', 'instant_bookable', 'cancellation_policy', 'room type',
       'Construction year', 'price', 'service fee', 'minimum nights',
       'number of reviews', 'last review', 'reviews per month',
       'review rate number', 'calculated host listings count',
       'availability 365', 'house_rules', 'license'],
      dtype='str')
```

In [ ]:
```
# Convert upper, lower, title, sentence case, etc.
for c in df3.columns:
    #print(c.upper())       # UPPERCASE
    #print(c.lower())       # lowercase
    #print(c.title())       # Title Case
    #print(c.capitalize())  # Sentence case
    #print(c.swapcase())    # Swap UPPER and lower
    print(c.casefold())    # Case-insensitive normalization
```

In [86]:
```
Capital_column_names = []
for C in df3.columns:
    C= C.upper()
    Capital_column_names.append(C)

Lower_column_names =[x.lower() for x in df3.columns]
```

In [95]:
```
df3.duplicated()
#df3.duplicated().sum()
#df3.duplicated().head()
#df3.duplicated().tail()
#df3.duplicated().value_counts()
```

Out[95]:
```
False    102058
True        541
Name: count, dtype: int64
```

In [100…
```
df3.drop_duplicates()
df3.duplicated().sum()
#df3.drop_duplicates(inplace=True)
#df3.duplicated().sum()
```

Out[100… np.int64(0)

Part 4 Remove Nan values from data

In [112…
```
df3.isna().sum()
#df3.isna().sum().sort_values(ascending=False)
```

Out[112…
```
id                                0
Name                            250
host id                           0
host_identity_verified          289
host name                       404
neighbourhood group              29
neighbourhood                    16
lat                               8
long                              8
country                         532
country code                    131
instant_bookable                105
cancellation_policy              76
room type                         0
Construction year               214
price                           247
service fee                     273
minimum nights                  400
number of reviews               183
reviews per month             15818
review rate number              319
calculated host listings count  319
availability 365                448
house_rules                   51842
dtype: int64
```

In [127…
```
#df3.drop(columns = ["license"], inplace=True)
#df3.columns
#df3.dropna()
df3.isna().sum()
```

Out[127…
```
id                                0
Name                              0
host id                           0
host_identity_verified            0
host name                         0
neighbourhood group               0
neighbourhood                     0
lat                               0
long                              0
country                           0
country code                      0
instant_bookable                  0
cancellation_policy               0
room type                         0
Construction year                 0
price                             0
service fee                       0
minimum nights                    0
number of reviews                 0
reviews per month                 0
review rate number                0
calculated host listings count    0
availability 365                  0
house_rules                       0
dtype: int64
```

In [114…
```
#df3.dropna(inplace = True)
#df3.isna().sum()
```

In [136…
```
df4 = data_orignal.copy()
print(df4["host_identity_verified"].dtype)
print(df4["host_identity_verified"].unique())
#df4["host_identity_verified"]
#df4["host_identity_verified"] = df4["host_identity_verified"].str.upper()
#df4["host_identity_verified"]
```

```
str
<StringArray>
['unconfirmed', 'verified', nan]
Length: 3, dtype: str
```

Part 5 Check more other transformations

In [135…
```
#df3.dropna(inplace = True)
print(df4["host_identity_verified"].dtype)
print(df4["host_identity_verified"].unique())
```

```
str
<StringArray>
['UNCONFIRMED', 'VERIFIED', nan]
Length: 3, dtype: str
```

In [137…
```
df4["instant_bookable"]
```

Out[137…
```
0         False
1         False
2          True
3          True
4         False
          ...
102594    False
102595     True
102596     True
102597     True
102598    False
Name: instant_bookable, Length: 102599, dtype: object
```

In [145…
```
df4["instant_bookable"].apply(lambda x : 1 if x== True else 0)
```

Out[145…
```
0         0
1         0
2         1
3         1
4         0
         ..
102594    0
102595    1
102596    1
102597    1
102598    0
Name: instant_bookable, Length: 102599, dtype: int64
```

In [149…
```
len(df4.columns)
```

Out[149…
```
26
```

In [150…
```
# Clean dollar sign from columns
```

In [152…
```
df4["price"][2]

#type(df4["price"][2]) # str
```

Out[152…
```
str
```

In [170…
```
df5 = data_orignal.copy()
df5["price"]
```

Out[170…
```
0          $966
1          $142
2          $620
3          $368
4          $204
          ...
102594     $844
102595     $837
102596     $988
102597     $546
102598    $1,032
Name: price, Length: 102599, dtype: str
```

In [174…
```
df5["price"] = df5["price"].str.replace("$","")

df5["price"]
```

Out[174…
```
0          966
1          142
2          620
3          368
4          204
          ...
102594     844
102595     837
102596     988
102597     546
102598    1,032
Name: price, Length: 102599, dtype: str
```

In [184…
```
df5["price"] = df5["price"].str.replace(",", "")
df5["price"]
```

Out[184…
```
0          966
1          142
2          620
3          368
4          204
          ...
102594     844
102595     837
102596     988
102597     546
102598    1032
Name: price, Length: 102599, dtype: str
```

In [189…
```
type(df5["price"][2])
df5["price"] = df5["price"].str.replace(" ","")
df5["price"].dropna()
```

Out[189…
```
0          966
1          142
2          620
3          368
4          204
          ...
102594     844
102595     837
102596     988
102597     546
102598    1032
Name: price, Length: 102352, dtype: str
```

In [195…
```
df5["price"].isna().sum()
df5.dropna(subset=["price"], inplace=True)
df5["price"]= df5["price"].astype(int)
```

In [ ]:
```
#df5.to_csv("Clean_data.csv")
#pd.read_csv("Clean_data.csv")
```

In [ ]:
```
#df5.to_csv("Clean_data_without_index.csv",index_label=False)
```

In [ ]:
```
#df5.to_excel("clean_data_excel.xlsx")
#pd.read_excel("clean_data_excel.xlsx")
```